MiniProjet-DeepLearning

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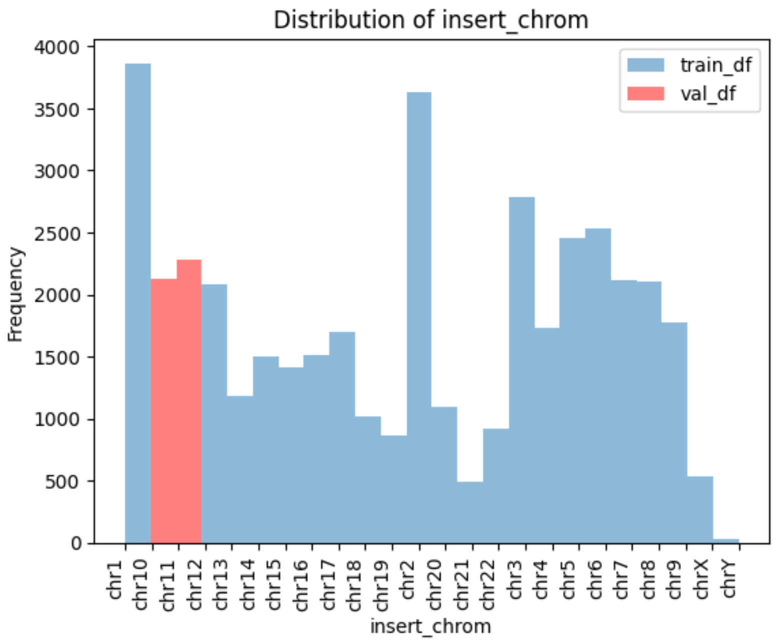
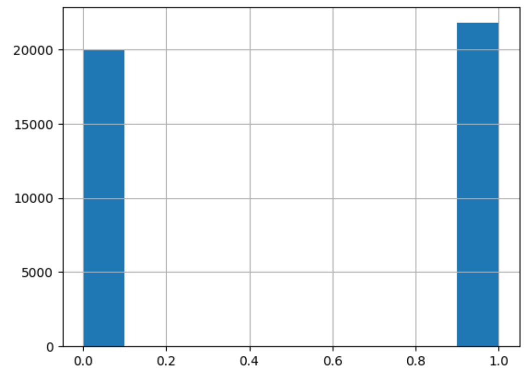
1 Data exploration

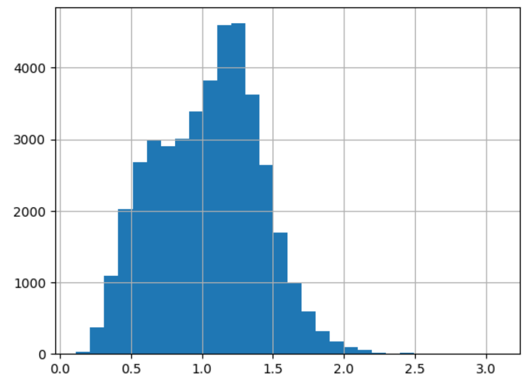
Analysis of the data showed that the binary variable `is_active` is very well balanced in our data set. The continuous `rna_dna_ratio`, on the other hand, seems to follow a Poisson-like distribution. We partitioned the data into training and validation sets by chromosome, a common strategy in genomics. From the full set of chromosomes, we then selected a small subset whose summary statistics most closely matched those of the full dataset. Finally, we used the performance on this validation set to design the architecture of our neural network.



Our choice of chromosomes for the architecture test



Kolumna `is_active` *peak*



kolumna `rna_dna_ratio`

Rysunek 2: The chromosomes we selected had a very similar distribution to the overall data.

Network Architecture

We designed a one-dimensional convolutional neural network (**SeqNN**) inspired by EfficientNet blocks, tailored for DNA sequence analysis. Key elements include:

- **Input Layer:** One-hot encoded sequences of length L with 4 channels (A, C, G, T), tensor $(B, 4, L)$. Ambiguous bases (N) are imputed randomly to preserve all samples.
- **Initial Convolution (blc0):** $Conv1d(4 \to 256,\ kernel = 5) + BatchNorm + SiLU$ for low-level motif extraction.

- **Inverted Residual Blocks with SE:** For each size pair (p, s) in [256,256,128,128,64,64,32,32]:

  1. $Conv1 \times 1(p \rightarrow s \times r)$, depthwise $Conv(p \times r)$, $SE_{Layer}(p, s \times r)$, $Conv1 \times 1(s \times r \rightarrow p)$
  2. Skip connection: concatenate input and SE output, then $Conv(p \times 2 \rightarrow s)$ to resize.

  Here $r = 4$ is the *expansion factor*. The `SE_Layer` uses Squeeze-and-Excite plus Bilinear to adaptively weight feature channels.

- **Feature Aggregation:** Adaptive max-pooling to fixed length ($pool_{dim} = 4$) reduces overfitting and handles variable sequence lengths.

- **Multi-task Heads:**
  - **Classification:** Fully connected layer on pooled features with Sigmoid activation for sequence activity probability.
  - **Regression:** Pointwise convolution ($Conv1 \times 1$ to $final_{channel} = 18$), flatten, then linear layer predicts RNA/DNA ratio.

Limitations

- **Pooling Compression:** Fixed $pool_d im$ can discard important positional information, affecting detection of long-range dependencies.

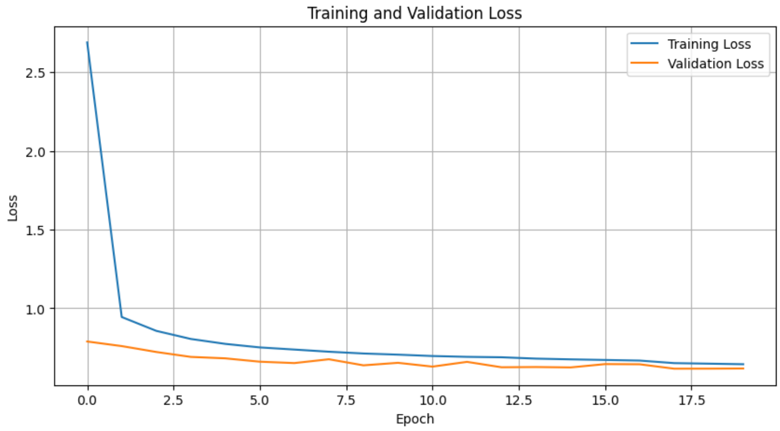
- **Low-Rank Bilinear Approximation:** Captures only pairwise interactions; higher-order dependencies might be missed.

- **Multi-task Trade-offs:** Simultaneous optimization of classification and regression can yield conflicting gradients.

- **Weak Binary Classification:** Poor classification of the binary variable `is_active`, much better predicts a continuous variable.

2 Final Experiment

Our final step was to perform a 5-fold cross-validation. Based on the highest F-score across the validation folds, we selected and saved our `best_model`, which also achieved a low mean squared error (MSE).



Rysunek 3: Run of the loss function (BCE + MSE) for the best model

3 Future Research Directions

- **Trainable Embeddings:** Replace one-hot with learned nucleotide embeddings or k-mer vectors.
- **Attention Mechanisms:** Explore Transformers for capturing long-range dependencies.
- **Dynamic Pooling:** Attention-based or content-adaptive pooling to preserve important motifs (Our experiments with ASSP gave worse results).
- **Baseline in Binary Classification:** Comparison of our method with standard machine learning methods such as AdaBoost or Random Forests, possibly another discrete variable prediction model. It is surprising to see a good result on a continuous variable and not the best result on a discrete variable, especially since you can predict whether a sequence is active based on the continuous variable (> 1.0473).